Introduction to Data Warehouses

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Data warehouse : is constructed by Integrating data from multiple heterogeneous sources that supports Analytical reporting and Decision making .

Data Warehousing involves :

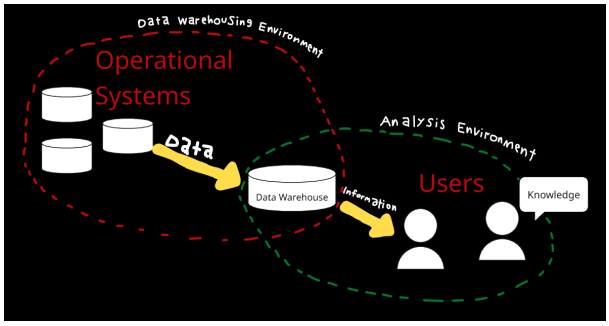
- Data Cleansing
- Data Integration
- Data Consolidations

1 Business Intelligence

Transformation **Data** to **Information**

Then

Derivation **Knowledge** from **Information**



2 Query Driven Approach

This is traditional approach to integrate heterogenous databases.

2.1 Disadvantages :

1. Need complex integration and filtering processes
2. Very inefficent
3. Very expensive for frequent queries

4. Very expensive for queries need to aggregations.

3 Update Driven Approach

Alternative to traditional approach , Must of modern data warehouses systems used this approach more than query-driven approach.

3.1 Advantages :

1. Provides high performance
2. data is copied, processed, integrated, annotated, summarized and restructured in semantic data store
3. Query processing does not require an interface to process data at local sources

4 OLAP vs OLTP

OLAP is **OnLine Analytical Processing** which refers to processing at warehouses

OLTP is **OnLine Transactions Processing** which refers to processing at operational systems 'databases' .

4.1 OLTP

1. Thousands of Users
2. Mostly updates
3. MBs or GBs of data
4. Raw Data
5. Many small transactions

4.2 OLAP

1. Hundreds of Users
2. Mostly reads
3. Provides a History
4. GBs or TBs of data
5. Queries are long and complex

5 Useful resources

The data warehousing institute: www.tdwi.org

Data warehousing on web :

- www.datawarehousing.org
- www.datawarehousing.com

Online Magazines and Periodicals :

- www.intelligententerprise.com
- www.dmgreview.com
- www.cio.com

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